

CoreTech Cybersecurity Internship Task

Introduction

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Reconnaissance Techniques in Ethical Hacking

Introduction

Reconnaissance is the process of gathering information about a target before performing a security assessment. It helps security professionals understand the target's systems, network, and online presence. Reconnaissance is divided into two types: Passive and Active.

1. Difference Between Passive and Active Reconnaissance

Passive Reconnaissance	Active Reconnaissance
Information is collected without directly interacting with the target.	Information is collected by directly interacting with the target.
Difficult to detect.	Can be detected by security systems.
Uses public sources.	Uses scanning and probing tools.
Lower risk.	Higher risk.
Examples: WHOIS, Google Search, DNS records.	Examples: Nmap scans, Port scanning, Vulnerability scanning.

Examples

Passive Reconnaissance

- WHOIS Lookup
- Google Search
- DNS Information Gathering
- Social Media Research

Active Reconnaissance

- Network Scanning
- Port Scanning
- Banner Grabbing
- Vulnerability Assessment

2. WHOIS Lookup Practice

Objective

Gather publicly available registration information about a domain.

Command

whois example.com

```
--(kali@kali)-[~]
└─$ whois example.com
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://res-dom.iana.org
Updated Date: 2026-01-16T18:26:50Z
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2026-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: ELLIOTT.NS.CLOUDFLARE.COM
Name Server: HERA.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 C988EC423E3880EB8D08A6FE06CA230EE23F35B578D64E78B29C3E1C83D245A
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-06-09T15:59:21Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
% IANA WHOIS server
% For more information on IANA, visit http://www.iana.org
% This query returned 1 object

domain:          EXAMPLE.COM

organisation:    Internet Assigned Numbers Authority

created:         1992-01-01
source:         IANA
```

Sample Findings

- Domain Name: example.com
- Registrar: IANA
- Creation Date: 1995
- Status: Active

Purpose

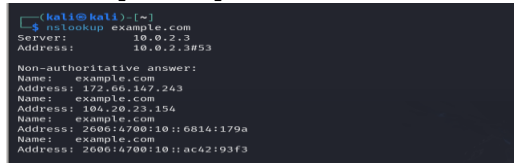
WHOIS helps identify:

- Domain owner information
- Registrar details
- Creation and expiration dates
- Name servers

3. DNS Information Gathering Using nslookup

Command

```
nslookup example.com
```



```
(kali@kali) [~]
$ nslookup example.com
Server:      10.0.2.3
Address:     10.0.2.3#53

Non-authoritative answer:
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 194.20.23.154
Name:   example.com
Address: 2606:4700:10::6814:179a
Name:   example.com
Address: 2606:4700:10::ac42:93f3
```

Sample Output

```
Server: 8.8.8.8
Address: 8.8.8.8
```

```
Name: example.com
Address: 93.184.216.34
```

Information Obtained

- Domain IP Address
- DNS Server Information
- Mail Exchange Records (MX)
- Name Servers

MX Record Lookup

```
nslookup -type=mx example.com
```



```
(kali@kali) [~]
$ nslookup -type=mx example.com
Server:      10.0.2.3
Address:     10.0.2.3#53

Non-authoritative answer:
example.com mail exchanger = 0 .

Authoritative answers can be found from:
```

4. DNS Information Gathering Using dig

Command

dig example.com

```
(kali@kali)-[~]
$ dig example.com

; <<>> DiG 9.20.0-Debian <<>> example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 9329
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                259     IN      A      104.20.23.154
example.com.                259     IN      A      172.66.147.243

;; Query time: 104 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Tue Jun 09 12:18:32 EDT 2026
;; MSG SIZE rcvd: 72
```

Useful Commands

A Record

dig example.com A

```
(kali@kali)-[~]
$ dig example.com A

; <<>> DiG 9.20.0-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 22645
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                239     IN      A      172.66.147.243
example.com.                239     IN      A      104.20.23.154

;; Query time: 188 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Tue Jun 09 12:00:45 EDT 2026
;; MSG SIZE rcvd: 61
```

MX Record

dig example.com MX

```
(kali@kali)-[~]
$ dig example.com MX

; <<>> DiG 9.20.0-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 39136
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                1       IN      MX      0 .

;; Query time: 136 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Tue Jun 09 12:01:10 EDT 2026
;; MSG SIZE rcvd: 55
```

NS Record

dig example.com NS

```
(kali@kali)-[~]
$ dig example.com NS

; <<> Dig 9.20.0-Debian <<> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 49409
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                 86400   IN      NS      hera.ns.cloudflare.com.
example.com.                 86400   IN      NS      elliottns.cloudflare.com.

;; Query time: 628 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Tue Jun 09 12:01:29 EDT 2026
;; MSG SIZE rcvd: 95
```

Information Obtained

- IP Addresses
 - Mail Servers
 - Name Servers
 - DNS Configuration Details
-

5. Google Dorking Techniques

Google Dorking uses advanced search operators to find publicly available information.

Example 1

```
site:example.com
```

Finds all indexed pages of a website.

Example 2

```
site:example.com filetype:pdf
```

Finds PDF files on a website.

Example 3

```
site:example.com intitle:index.of
```

Finds directory listings.

Example 4

```
site:example.com inurl:login
```

Finds login pages.

Example 5

```
site:example.com filetype:doc
```

Finds Word documents.

Important Note

Google Dorking should only be used for educational and authorized security assessments. Do not attempt to access private or restricted information.

6. Shodan Search Concepts

What is Shodan?

Shodan is a search engine that finds internet-connected devices such as:

- Servers
- Routers
- Webcams
- IoT Devices
- Industrial Control Systems

Example Searches

Web Servers

```
apache
```

SSH Servers

```
port:22
```

HTTPS Services

```
port:443
```

Devices in Pakistan

```
country:PK
```

Microsoft IIS Servers

```
product:"Microsoft-IIS"
```

Information Obtained

- Open Ports
- Running Services
- Software Versions
- Device Locations

Ethical Use

Only observe publicly available information and never attempt unauthorized access.

Findings Summary

Tool	Purpose	Information Gathered
WHOIS	Domain Registration	Owner, Registrar, Dates
nslookup	DNS Query	IP Address, DNS Records
dig	Advanced DNS Query	A, MX, NS Records
Google Dorking	Search Engine Recon	Publicly Indexed Information
Shodan	Internet Device Search	Open Ports, Services